ENAE 464. Aerospace Engineering Laboratory
Lab 6. Particle Imaging Velocimetry (PIV)

Date/Time: 4/23/20 Th 2-5pm (Section 0201)
4/24/20 F 2-5pm (Section 0202)

Place: Thermal-Fluids Instructional Lab (KIM 1209)

1. Introduction

Particle Imaging Velocimetry (PIV) is an optical diagnostic technique that is used to measure an instantaneous velocity field for a cross-sectional plane in the flow field. The basic principle of PIV is to measure the spatial displacement of a fluid element, illuminated by two short-exposure light sheets with a known time interval. By making the time interval sufficiently short, an instantaneous velocity vector can be approximated. Also, by tracking many fluid elements at the same time, the velocity vector field can be calculated.

The typical approach for PIV is given as follows:

1) The flow is seeded with particles (called seed or tracer particles) that are sufficiently small to accurately follow the fluid flow.
2) Seed particles (in a plane along the flow direction) are then illuminated by a thin light sheet generated by a double-pulsed laser.
3) The laser light scattered by the seed particles is imaged with a CCD camera, which is placed perpendicular to the laser sheet plane.
4) Two separate images corresponding to each double-pulse laser light source are obtained.
5) The displacement (Δx) of the seed particles that are carried by the fluid during the given time delay (Δt, between the double pulses) is estimated with a cross-correlation technique that analyzes particle-pattern in small sub-domain, or called interrogation window, between the two images. Then the fluid velocity is determined by $u = \Delta x / \Delta t$.

In this experiment, we will measure the downstream wake of a circular cylinder in a water tunnel in the Thermal-Fluids Instructional Lab to demonstrate the PIV technique for flow visualization. The water tunnel test section is 6-in high, 6-in wide and 12-in long. A cylinder of 0.5-in diameter is placed horizontally at the middle height of the test section and covers the entire width of the test section. The water flow speed in the tunnel is controlled by the speed of the pump.

2. PIV Experimental Setup

In this experiment, we use a PIV system from LaVision Inc. The system includes a CCD camera, a double-pulsed Nd:YAG laser and an PIV software for image processing (see Figure 1). In addition, the particle seeding and light sheet formation are described as follows.

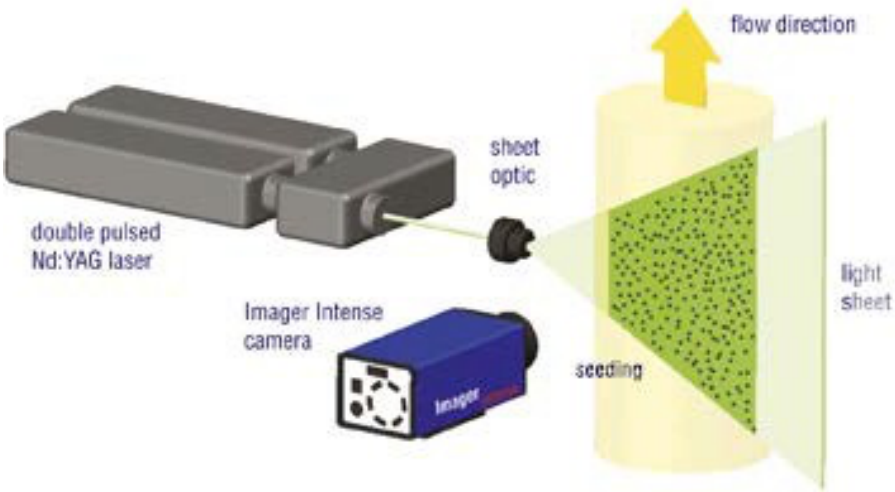


Figure 1. PIV setup

Laser light sheet formation: The light source for 2D PIV is required to be a thin light sheet with a short duration of illumination. The thin light sheet is to make the images in focus and the short duration of illumination is to 'freeze' the motion of the particles during pulse exposure in order to prevent image blurring. Lasers that produce a pulsed and collimated light beam can easily fulfill these requirements. In the current experiment, a circular laser beam from an Nd:YAG laser with a wavelength of 532 nm is shaped into a thin sheet by means of cylindrical lenses. The light sheet is vertical along the center of water tunnel. The thickness of the light sheet in measurement region is between 100µm and 1 mm. The laser sheet pulse and camera shutter are synchronized with LaVision software.

Particle seeding: As discussed, seed particles are required for laser scattering and they need to be small enough to follow the fluid. The scattering ability of particles depends on the size of the particles as well as the index of refraction relative to the fluid. Thus, the seed particles also need to be sufficiently large to scatter light efficiently so that they are clearly detected through imaging devices. In this experiment, polyamide particles of an average diameter 20 µm and density 950 kg/m^3 are used. The particles are randomly distributed in the water flow, and the seeding concentration is estimated to be between 10^9 and 10^{12} particles/m^3. An optimal seeding concentration for the PIV software we will use is about 7 to 10 particles per the interrogation region.

3. Experimental Procedure

- Operate the water tunnel at a specified setting. The frequency setting for controlling the water flow varies from 2.0 to 2.5 Hz. Each group will be assigned one frequency, which controls the water pump by establishing a steady flow with the average speed such that $U_\infty = f$(pump frequency).
- Operate the LaVision system under supervision (**Safety issue: Everybody in group should wear laser safety goggle**).

- Using three different time delays, obtain three separate sets of image sequences.
- Using the PIV software, conduct image processing. The interrogation window size is 32 by 32 pixels and vector spacing is 16 pixels. The calibration scale is done before the class.
- Save your results.

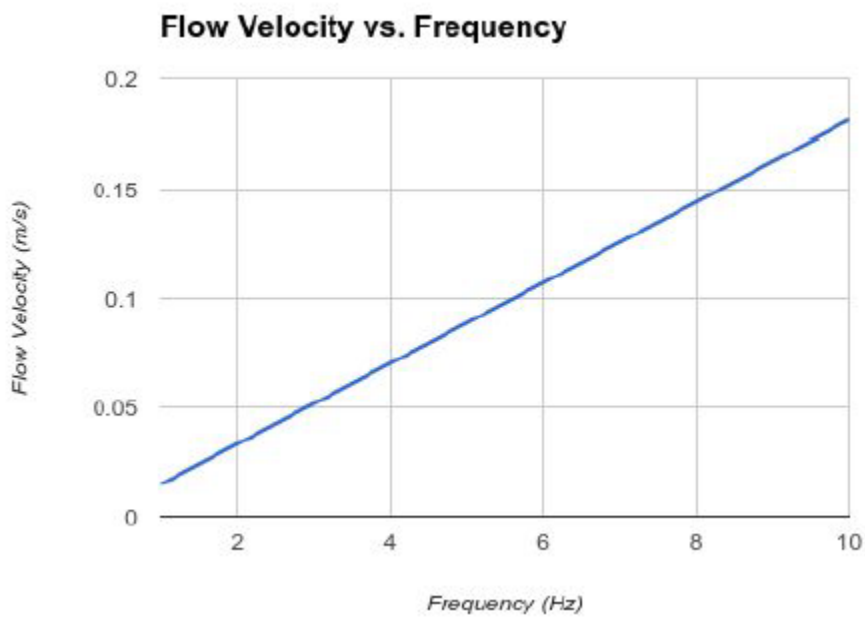
4. Reported Information

The reported information should contain the following items and include figures and calculations:

1. Measured flow velocity and Reynolds number – Calculations of the incoming water flow speed ($\overline{U}_\infty$) and the appropriate Reynolds number for the water flow over the cylinder.
2. Expected shedding frequency and Strouhal number – It is well known that a cylinder immersed in a steady flow produces a periodic vortex shedding, referred to as a Kármán vortex street. Look up the preferred Strouhal number for your case, and from this, estimate the shedding frequency in Hz.
3. Stokes number for visualization – Using the shedding frequency as the inverse of the characteristic flow time scale (i.e., $f_{shedding}=U/L$), calculate the Stokes number for the particle seeding.
4. Plot y-velocity fluctuation in the vicinity of (x,y)=(0,0) as a function of time. From this, fit a sinusoid to estimate the measured vortex shedding frequency and compare this with the expected value in part 2. (These should be close, i.e. not ~order of magnitude different).

Notes/measurements needed:

- The pump frequency is called out in the data file title, e.g. 2p5hz is 2.5 Hz.



- The relationship between flow velocity and pump frequency is linear, given as:

$$U_\infty(Hz) = 0.0185 \times Hz - 0.0038$$

- Diameter of the cylinder is 0.0127 m.
- The Strouhal number can be found in references online for the case of a cylinder experiencing cross flow.

The Stokes number for a flow with suspended particles is the ratio of the typical reaction time of the particles in the flow to the characteristic flow time. The former is a function of the particle size and density and is essentially a characterization of the ability of the flow to force it to move. The latter is representative of the timescales that the flow varies on. For this case it is essentially the time it takes the flow to pass the cylindrical obstacle.

The particle reaction time is given by:

$$T_V = \frac{\rho D^2}{18\eta}$$

where ρ is the particle density (950 kg/m^3) for this experiment, D is the particle diameter, 20 microns, and η is the dynamic flow viscosity: $4.04*10^{-4}$ kg/(m-s).

- The data files are formatted strangely; you can create sensible vectors for (x,y) position and (u,v) velocity using this MATLAB code snippet:

```
% Compensate for weird format with reshape
x(i,:) = reshape((data(1:120,1:67)), 1, []);
y(i,:) = reshape((data(121:240,1:67)), 1, []);
u(i,:) = reshape((data(241:360,1:67)), 1, []);
v(i,:) = reshape((data(361:480,1:67)), 1, []);
```

- The total test duration is 0.38 seconds.
- Use MATLAB's 'quiver' command to plot a vector flow field.